Computer Assignment 1 Report

MATH 105LA

Rye Mauhay

Problem 1

Use fixed-point iteration to find a fixed point of $g(x) = x/2 + 1/x$ in $[1, 2]$ to approximate $\sqrt{2}$ accurately to within 10^{-5}. Find a root of $f(x) = x - g(x)$ using Newton's method under the same tolerance level. Which method converges faster?

Solution

Method

For fixed-point iteration, we observe that

$$g([1, 2]) = [\sqrt{2}, 1.5] \subset [1, 2]$$

and therefore there must exist a fixed point in the interval [1,2]. We employ fixed-point iteration with $a_0 = 1$. Such algorithm will repeatedly apply the function g on successive points until $|p_n - p_{n-1}| \leq 10^{-5}$ for some n.

For Newton's method, we utilize the function

$$f(x) := x - g(x)$$

for which the first derivative is simply

$$f'(x) = 1 - g'(x) = 1 - \frac{1}{2} - (-\frac{1}{x^2}) = \frac{1}{2} + \frac{1}{x^2}$$

We will also set $a_0 = 2$.

Results

The fixed-point iteration method successfully terminated with $n = 4$ iterations. The final approximation (to 6 decimal places) is

$$p^* \approx 1.414214$$

where

$$g(p^*) \approx 1.4142$$

The rate of convergence was calculated to be

$$\text{rate} \approx 1.999509$$

Newton's method successfully terminated with $n = 3$ iterations. The final approximation (to 6 decimal places) is

$$p^* \approx 1.414211$$

where

$$|f(p^*)| \approx 2.1239 \times 10^{-6}$$

The rate of convergence was calculated to be

$$\text{rate} \approx 1.620263$$

Figure 1 depicts every iterated point from the fixed-point iteration on the horizontal axis. The vertical axis depicts its actual error.
Figure 2 depicts every x-intercept from Newton's method on the horizontal axis. The vertical axis depicts its actual error.

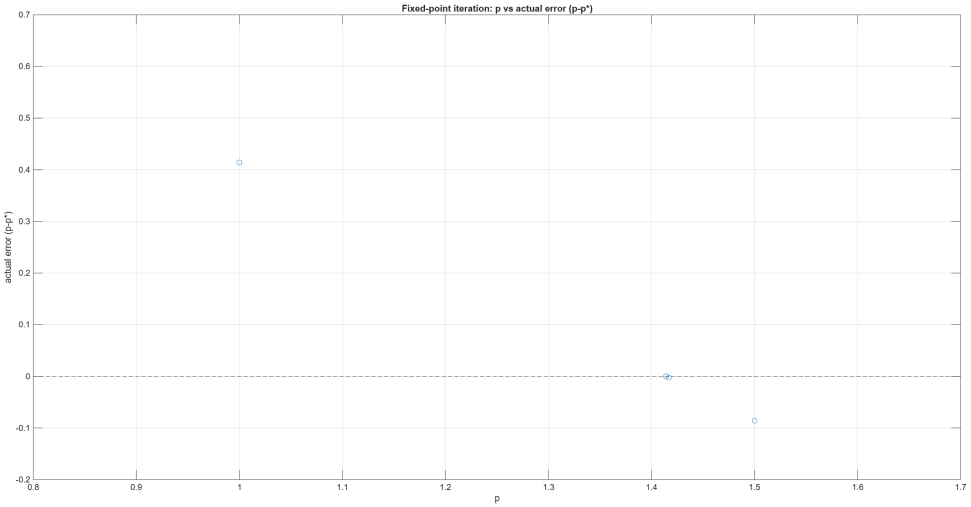


Figure 1: Fixed-point iteration results

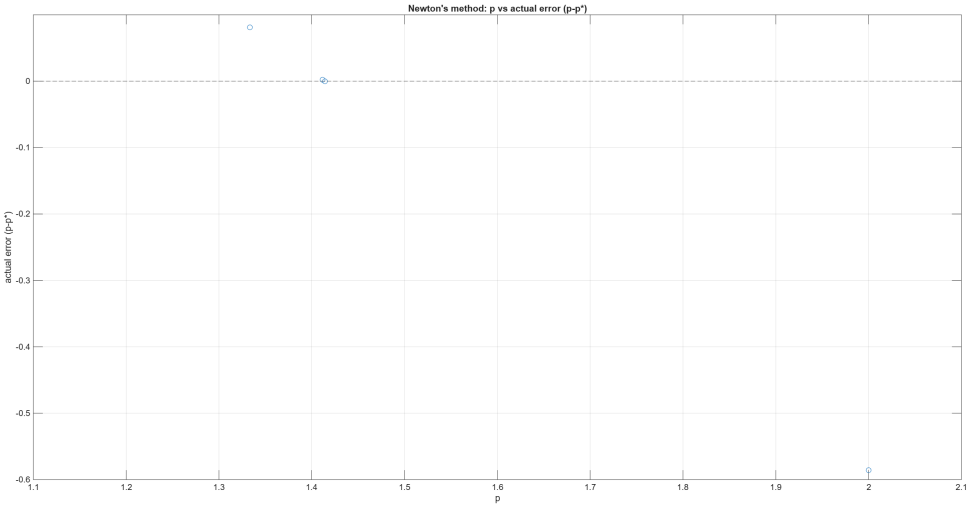


Figure 2: Newton's method results

Problem 2

An object falling vertically through the air is subjected to viscous resistance as well as to the force of gravity. Assume that an object with mass m dropped from a height s_0 so that the height of the object after t seconds is

$$s(t) = s_0 - \frac{mg}{k}t + \frac{m^2 g}{k^2}\left(1 - e^{-kt/m}\right), \tag{1}$$

where $g = 32.17$ ft/s^2 and k represents the coefficient of air resistance in lb-s/ft. Suppose $s_0 = 300$ ft, $m = 0.25$ lb, and $k = 0.1$ lb-s/ft. Find, to within 0.01 second, the time it takes this quarter-pounder to hit the ground.

Solution

Method

We utilize the bisection method because calculating the derivative of $s(t)$ for Newton's method may be too complex. We observe that

$$s(0) = 300, s(10) \approx -306.8701$$

Since they have different signs on the points $\{0, 10\}$ and $s(t)$ is continuous on the interval $[0, 10]$, we are guaranteed a root by the Intermediate Value Theorem. We use a tolerance of 10^{-3} as prescribed by the problem.

Results

The bisection method successfully terminated with $n = 10$ iterations. The final approximation (to 6 decimal places) is

$$p^* \approx 6.000977 \text{ seconds}$$

where

$$|s(p^*)| \approx 0.2011$$

Figure 3 depicts midpoint iterated through in the bisection method on the horizontal axis. The vertical axis depicts the actual error.

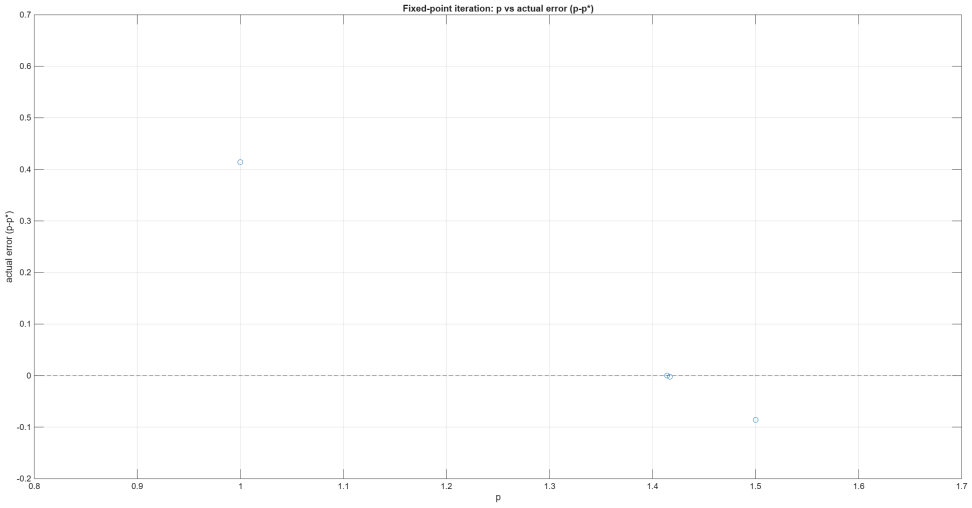


Figure 3: Bisection method iteration results

Discussion

Though Newton's method had the least number of iterations ($n = 3$), it had a smaller rate of convergence than the fixed-point iteration. Additionally, using Newton's method requires calculation of the derivative, which may be computationally exhaustive.